

Why the squarefree prefix Gram matrix does not force factorial rank

Asymptotic method diagnostic; independently reviewed by all three collaborators

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Normalize each letter's total mass to one. A positive weighted word is viewed as an axis path with cumulative coordinates $X_i(t)$; the coordinate X_i increases only on the segments carrying letter i . For a word u on distinct letters, write $F_u(t)$ for the corresponding prefix iterated integral. Put $F_\emptyset = 1$. The endpoint signature of a fair family has coefficient $1/k!$ on every word of k distinct letters.

Theorem 1 (A low-rank positive completion). *Fix a marked letter c and an integer $r \geq 1$. Assume fairness through degree $2r + 1$. Index prefix functions by all distinct-letter words u of length at most r avoiding c , and consider their Gram matrix in $L^2(dX_c)$. Fairness and the shuffle identity determine, whenever the supports of u, v are disjoint,*

$$G_{uv} := \int F_u(t) F_v(t) dX_c(t) = \frac{1}{|u|! |v|! (|u| + |v| + 1)!}. \quad (1)$$

These specified entries admit a positive semidefinite completion of rank at most $r+1$, regardless of the size of the alphabet. Restricted to words of a single fixed degree, they admit a rank-one positive semidefinite completion.

Proof. For disjoint supports, the shuffle product contains $\binom{a+b}{a}$ distinct interleavings, where $a = |u|$, $b = |v|$. Appending the marked letter makes every resulting endpoint word have $a + b + 1$ distinct letters. Therefore

$$\int F_u F_v dX_c = \sum_{w \in u \text{ shuffle } v} \text{Sig}_{wc} = \frac{\binom{a+b}{a}}{(a+b+1)!},$$

which is (1). Here u shuffle v simply denotes the set of order-preserving interleavings.

Now set the right side of (1) as the value of *every* matrix entry, including pairs with overlapping supports. This is the Gram matrix on $[0, 1]$ of the functions

$$f_u(t) = t^{|u|} / |u|!.$$

It is positive semidefinite and its functions span a space of dimension at most $r + 1$. At fixed degree all the functions coincide, so that block has rank one. This completion is an algebraic compatibility statement, not a claim that every finite fair word has those particular repeated-letter moments. $\square$

What disjoint polarization actually says. For two orderings u, u' of one support, put $D = F_u - F_{u'}$. Equation (1) says that D is orthogonal to every disjoint-support prefix function covered by the degree bound. Two such order contrasts on disjoint supports are also orthogonal. Their squared norms, however, involve repeated letters and are not specified by (1). In the positive completion above every contrast is identically zero. Thus disjoint

polarization and Gram positivity alone do not force even nonzero norms for these contrasts, much less factorially many independent prefix functions. Adding the mean moments (empty word) does not repair the argument: they are already included in the same rank- $(r + 1)$ completion.

Proposition 1 (The continuous diagonal realization). *The completion in Theorem 1 is realized by the continuous positive path $X_1(t) = \dots = X_n(t) = t$, $0 \leq t \leq 1$. This path is exactly fair in every squarefree degree. Its all-time endpoint insertion directions, projected through Lie degree $r \leq n$, span a space of dimension at most*

$$n + (r - 1)(n - 1).$$

Proof. Every prefix iterated integral of degree a is $t^a/a!$, and the endpoint signature is $\exp(H)$, where $H = \sum_i X_i$ is now the sum of the formal Lie generators. The insertion directions are

$$\text{Ad}_{\exp(tH)} X_i = \sum_{a=0}^{r-1} \frac{t^a}{a!} (\text{ad } H)^a X_i \quad \text{through degree } r.$$

The degree-zero-in- t vectors span at most n dimensions. For each $a \geq 1$, the n vectors $(\text{ad } H)^a X_i$ have the relation $\sum_i (\text{ad } H)^a X_i = (\text{ad } H)^a H = 0$, so they span at most $n - 1$ dimensions. Summing these bounds proves the assertion. $\square$

This continuous example is *not* a finite axis-word construction. It shows precisely why a factorial-rank proof must exploit the restriction that only one coordinate moves at a time. Endpoint fairness, nonnegative coordinates, shuffle identities, and positivity of prefix Gram matrices remain compatible with the much smaller rank in Proposition 1.

Why deleting repeated letters is not a positive repair. One cannot simply set all repeated-letter products to zero inside a positive moment argument. In the commutative reduced shuffle algebra, a nonempty monomial a has $a^2 = 0$. A unital positive functional would have

$$0 \leq L((1 + ta)^2) = 1 + 2tL(a) \quad (t \in \mathbb{R}),$$

forcing $L(a) = 0$. In particular this contradicts the nonzero first-letter moments of the fair target. Equivalently, the proposed moment matrix on $1, a$ would have a nonzero off-diagonal entry and a zero diagonal entry, hence would fail positivity. Repeated-letter entries must therefore be supplied rather than discarded; the low-rank completion above is one legitimate supply at the level of these moment constraints.

Finite words and the remaining question. There are separate genuine finite-word obstructions to full automatic regularity. The reviewed note `../size_bounds/all_gap_singularity_inheritance.tex` constructs arbitrarily large fully fair positive words whose all-gap endpoint map is not surjective even through a fixed truncation degree $r \geq 5$. Its fixed singular seed is used only to falsify an asymptotic universal lemma. The present argument is independent of that seed and of its arithmetic certificate.

Neither result rules out a weaker factorial-order lower bound on the rank of every finite fair word. Such a theorem would still be a substantial route to a stronger total-size lower bound. It needs additional information from the actual axis-word geometry or from repeated-letter integrals; the squarefree Gram and disjoint-polarization identities on their own cannot supply it. The separate standard-insertion prefix collapse in `../independent_directions/prefix_signature_rank_obstruction.tex` provides an exact finite-word warning against raw prefix independence.